CO4 – AT1 - Article Review

5G / 6G Communication Systems

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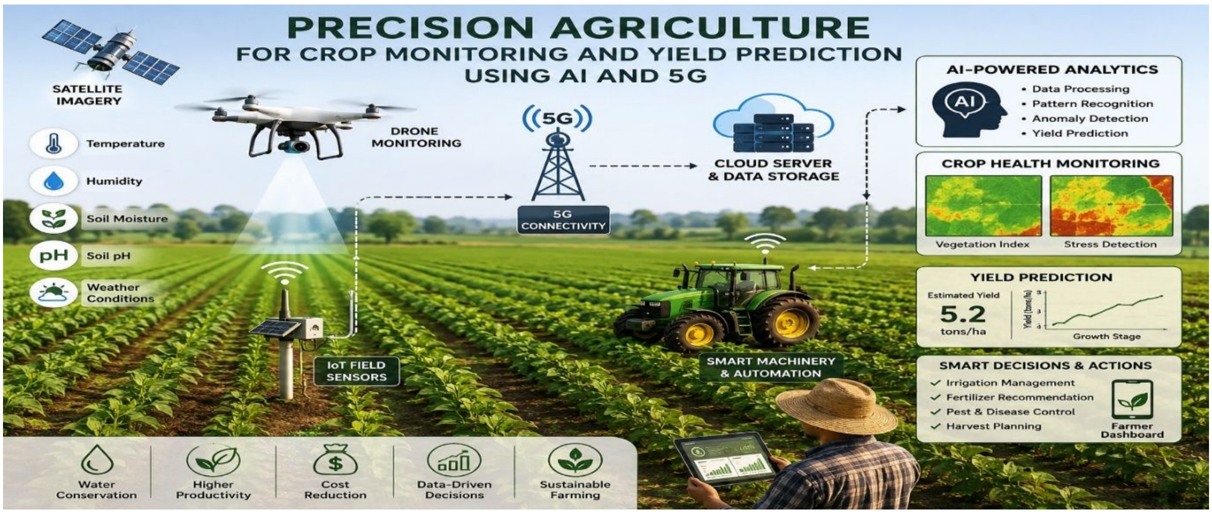
Topic: ***Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G***



Abstract

Precision agriculture, facilitated by advancements in Artificial Intelligence (AI), has emerged as a transformative paradigm in modern farming. This review comprehensively examines the integration of AI technologies in precision agriculture to enhance sustainability and optimize farming practices. The paper synthesizes recent research and developments in AI applications, covering key areas such as crop monitoring, resource management, decision support systems, and automation. The adoption of AI-driven techniques, including machine learning, computer vision, and sensor technologies, is reshaping traditional farming methods by providing farmers with real-time data and actionable insights. Crop monitoring applications utilize satellite imagery, drones, and ground-based sensors to assess plant health, detect diseases, and optimize irrigation strategies. AI-driven decision support systems empower farmers to make informed choices based on data-driven predictions, weather forecasts, and historical patterns, contributing to resource efficient practices and minimizing environmental impact. Resource management is a critical aspect of sustainable farming, and AI plays a pivotal role in optimizing the use of water, fertilizers, and pesticides. Smart irrigation systems, enabled by AI

algorithms, ensure precise and efficient water distribution, reducing water wastage and promoting water conservation.

1. Introduction

Precision agriculture is a modern farming approach that uses sensors, IoT devices, satellite/drone images, artificial intelligence (AI), and communication technologies to monitor crops and improve farm management. The main objective is to increase crop productivity while reducing the unnecessary use of water, fertilizers, pesticides, and other resources.Recent research shows that AI and machine learning can analyze agricultural data such as soil moisture, temperature, humidity, rainfall, nutrient levels, crop images, vegetation indices, and historical yield to estimate crop health and future yield. (UK Scientific Publishing)The addition of 5G communication can support this system by providing high-speed, low-latency connectivity between field sensors, drones, edge-computing devices, and cloud platforms. This makes near-real-time crop monitoring and decision-making more practical.

Keywords: Precision agriculture; Artificial Intelligence (AI); Sustainable farming; Technology review; Crop monitoring

2. Literature / Article Review

Author / Year	Main Focus	Technology Used	Major Findings
Saha et al. (2025)	Precision agriculture and crop-yield prediction	AI, ML, DL, Remote Sensing	Precision agriculture can improve crop-yield prediction, productivity, and resource management. Frontiers
Jabed & Murad (2024)	Crop-yield prediction	Machine Learning, Deep Learning	AI-based models can analyze complex agricultural factors and improve yield estimation. ScienceDirect
Crop Yield Prediction Review (2025)	ML-based crop-yield prediction	ML, Remote Sensing, Environmental Data	Machine learning is increasingly used for accurate yield forecasting and better agricultural decision-making. ScienceDirect
Mohan et al. (2025)	AI and Explainable AI for crop yield	AI, XAI, Decision Tree, LightGBM	Explainable AI helps identify the factors influencing crop-yield predictions and improves model interpretability. Frontiers
Mohammed et al. (2025)	AI for sustainable precision agriculture	AI, ML, Computer Vision, IoT, Robotics	AI supports crop monitoring, irrigation, pest/weed control, yield prediction, and real-time decision-making. FAO AGRIS
AI Crop-Yield Review (2025)	AI-based yield estimation	AI, ML, DL	AI models are increasingly being studied for accurate and sustainable crop-yield estimation. MDPI

3. Article 1 – Precision Agriculture for Improving Crop Yield Predictions

Saha et al. reviewed the use of precision agriculture technologies for improving crop-yield prediction. The study explains that precision agriculture collects field-specific information and uses it to optimize crop production, reduce operating costs, and improve sustainability. (Frontiers)

Important points:

- Uses data-driven farming techniques.
- Combines crop, soil, environmental, and field information.

- Helps predict crop productivity.
- Supports efficient use of agricultural resources.
- Reduces unnecessary inputs.
- Improves decision-making for farmers.

Limitation: Prediction performance can vary depending on crop type, environmental conditions, data quality, and geographical location.

4. Article 2 – Machine Learning for Crop Yield Prediction

Jabed and Murad presented a comprehensive review of machine-learning and deep-learning approaches for crop-yield prediction. Their work highlights the importance of AI for food production planning and resource optimization. (PubMed Central (PMC))

Commonly investigated techniques include:

- Linear Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- Artificial Neural Network
- Convolutional Neural Network
- Recurrent Neural Network
- Deep Learning models

AI models can learn relationships between environmental factors and historical crop yields. Once trained, these models can estimate future yield from new field data.

Limitation: AI models require sufficient and reliable historical datasets. Poor-quality or geographically limited data can reduce prediction accuracy.

5. Article 3 – AI Applications in Precision Agriculture

Gupta and Pal reviewed AI applications in precision agriculture, particularly crop monitoring, pest management, and resource optimization. (Springer)

AI can analyze:

- Crop images
- Leaf conditions
- Soil parameters

- Weather information
- Irrigation requirements
- Pest and disease symptoms
- Crop growth patterns

Computer vision can detect visual abnormalities in plants, while machine-learning algorithms can classify crop conditions and provide recommendations.

Main contribution: AI changes agriculture from traditional observation-based farming toward data-driven decision-making.

6. Article 4 – AI and Explainable AI for Crop Yield Prediction

Jagan Mohan and colleagues investigated the integration of AI and Explainable AI (XAI) for precision crop-yield prediction. (Frontiers)

Traditional AI models may produce a prediction without clearly explaining the reason behind it. XAI helps identify important factors affecting the prediction.

For example, an AI model may indicate that:

Low soil moisture + high temperature + low rainfall → reduced expected yield.

This is useful because farmers can understand the reason for an AI recommendation rather than simply receiving a prediction.

7. Article 5 – Recent Systematic Review and Future Research

A 2026 systematic review covering research from 2020–2025 identifies multimodal data fusion, real-time data assimilation, and hybrid physics/AI models as important directions for crop-yield prediction. It also identifies challenges such as model transferability, lack of standardized datasets, real-time integration, and explainability. (UK Scientific Publishing)

This is particularly relevant to a proposed AI + 5G precision agriculture system, because different data sources can be combined:

Sensors + Drone Images + Satellite Data + Weather Data + Historical Yield → AI Model → Crop Monitoring + Yield Prediction

8. Role of 5G in Precision Agriculture

5G can act as the communication layer connecting agricultural devices.

Basic architecture

Field Sensors

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5G Network

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Edge/Cloud Computing

↓

AI/ML Model

↓

Crop Health & Yield Prediction

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Farmer Dashboard / Mobile Alert

Advantages of 5G

1. High-speed communication – large sensor and image datasets can be transmitted quickly.
2. Low latency – supports fast responses to field conditions.
3. Large device connectivity – many IoT sensors can be connected.
4. Drone connectivity – enables transmission of crop images/video.
5. Edge AI support – AI processing can be performed closer to the farm.
6. Real-time monitoring – farmers can receive crop alerts quickly.

9. Proposed System Based on the Literature

Based on the reviewed research, the proposed system can combine AI, IoT, 5G, and precision agriculture.

Input parameters

- Soil moisture
- Soil temperature
- Air temperature
- Humidity
- Rainfall
- Soil nutrients
- Crop images
- Satellite/drone data

- Historical crop yield

Processing

The sensors continuously collect agricultural data. The 5G network transfers the data to an edge/cloud platform. An AI model analyzes the collected information.

Output

The system can provide:

- Crop health status
- Disease/stress indication
- Irrigation requirement
- Fertilizer requirement
- Expected crop yield
- Early warning alerts
- Field productivity map

10. Research Gap

From the reviewed literature, several research gaps can be identified:

- Many systems focus only on AI or IoT, rather than integrating AI with high-speed communication.
- Different agricultural data sources are not always combined effectively.
- AI models may not generalize well to different crops and geographical regions.
- Real-time agricultural datasets are limited.
- Some deep-learning models are difficult for farmers to interpret.
- Rural connectivity and deployment cost remain practical challenges.
- Standardized datasets for evaluating crop-yield prediction models are still needed. (UK Scientific Publishing)

11. Future Scope

Future systems can integrate:

- 5G + Edge AI
- Drones and autonomous agricultural robots
- Satellite remote sensing

- Digital twins of farms
- Explainable AI
- Multimodal AI
- Smart irrigation
- Automated fertilizer application
- Autonomous harvesting
- Predictive disease detection

Recent research particularly points toward multimodal data fusion and scalable edge/cloud architectures as promising directions. (UK Scientific Publishing)

Conclusion

The literature indicates that AI significantly improves crop monitoring and yield prediction, while IoT provides continuous field data. 5G can strengthen this architecture by providing fast and low-latency communication between sensors, drones, edge devices, and cloud platforms. Therefore, integrating Precision Agriculture + AI + IoT + 5G can create an intelligent farming system capable of monitoring crop conditions, predicting yield, and providing timely recommendations to farmers.

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